CHEMISTRY SOLUTIONS

1. C) $C_6H_{12}O_6$ | Explanation : The empirical mass is 30 amu; 180/30 = 6, so the molecular formula is $C_6H_{12}O_6$.

2. B) N_2 | Explanation : 3.011×10^{23} molecules = 0.5 mol. If 0.5 mol weighs 14 g, molar mass = 28 g/mol, which is N_2.

3. C) 6 | **Explanation:** For n=2n=2n=2 and l=1l=1l=1, it is the 2p subshell which contains 3 orbitals. Each orbital can hold 2 electrons, so total electrons 3×2=6.

4. B) $N^{3-} > O^{2-} > F^{-} > Na^{+} > Mg^{2+}$ | Explanation : In an isoelectronic series, ionic radius decreases with increasing nuclear charge.

5. C) 16 | **Explanation:** The total number of orbitals in a shell is given by n^2. For the 4th shell (n=4)(n=4)(n=4), the number of orbitals = 4^2=16.

6. B) +6

7. B) It halves. | Explanation : Pressure is inversely proportional to volume according to Boyle's law.

8. B) 8 | **Explanation:** In a BCC lattice, each atom is surrounded by 8 nearest neighboring atoms. Therefore, its coordination number is 8.

9. D) It is unitless. | Explanation : $\Delta n = 0$, so pressure units cancel completely.

10. D) Exactly neutral | Explanation : Acid and base milli-equivalents are equal (5 mEq each).

11. B) 11.3 | Explanation : $[OH^-] = 2\times10^{-3}$ M, pOH = 2.7, therefore pH = 11.3.

12. A) 1.0×10^{-4} M | Explanation : $Ksp = 4s^3 = 4\times10^{-12}$, giving $s = 10^{-4}$ M.

13. B) 0.693 / k | Explanation : For first-order reactions, $t_{1/2} = 0.693/k$.

14. C) 16 times | Explanation : A 40°C rise means four 10°C intervals, so rate increases by $2^4 = 16$.

15. C) 5 Faradays | Explanation : Mn changes from +7 to +2, requiring 5 electrons per mole.

16. A) Above 300 K | Explanation : Spontaneity begins when $T > \Delta H/\Delta S = 300$ K.

17. C) 7/8 | Explanation : After 3 half-lives, 1/8 remains, so 7/8 has decayed.

18. C) Mercury (Hg) | Explanation : Mercury toxicity, particularly from methylmercury accumulating in the food chain, causes Minamata disease, resulting in severe damage to the central nervous system.

19. C) Graphite | Explanation : In graphite, each carbon atom is sp^2 hybridized, bonded to three other carbons in a planar ring. The fourth valence electron is delocalized across the layer, allowing electrical conductivity.

20. B) Carbonate and hydrated oxide ores in the absence or limited supply of air. | Explanation : Calcination involves heating an ore below its melting point in the absence of air to expel volatile impurities, water of crystallization, or carbon dioxide (e.g., converting metal carbonates to metal oxides).

21. C) Charring | Explanation : Concentrated H_2SO_4 strips the elements of water (H and O in a 2:1 ratio) from carbohydrates like sucrose, leaving behind elemental carbon in a reaction known as charring.

22. D) $ZnSO_4.7H_2O$ | Explanation : Zinc sulfate heptahydrate ($ZnSO_4.7H_2O$) is historically and commercially known as white vitriol. $CuSO_4$-$5H_2O$ is blue vitriol, and $FeSO_4.7H_2O$ is green vitriol.

23. C) By coupling the inward 'downhill' transport of Na^+ with the inward 'uphill' transport of glucose. | Explanation : The sodium-glucose pump utilizes secondary active transport. The primary Na^+/K^+ pump uses ATP to create a high extracellular Na^+ gradient. The SGLT then uses the energy from Na+ flowing back down its gradient into the cell to simultaneously drag glucose into the cell against its gradient.

24. B) The crystal field splitting energy (Ao) is greater than the pairing energy (P), resulting in forced electron pairing in the t2g orbitals (low-spin complex). | Explanation : Strong field ligands cause a large splitting of the d-orbitals. Because Ao > P, it takes less energy for electrons to pair up in the lower t2g orbitals than to jump to the higher eg orbitals, creating a low-spin complex.

25. C) To rapidly oxidize and remove impurities like carbon, silicon, and phosphorus as volatile oxides or removable slag. | Explanation : The injected oxygen reacts exothermically with the impurities in the pig iron (C, Si, P, Mn), converting them into gases (like CO) or solid oxides that react with basic fluxes (like lime) to form a removable slag layer, thus purifying the iron into steel.

26. C) Calomel: Hg_2Cl_2 (+1), Corrosive Sublimate: $HgCl_2$ (+2) | Explanation : Calomel is mercurous chloride (Hg_2Cl_2), a relatively harmless compound where mercury is in the +1 oxidation state (existing as the diatomic cation Hg_2^{2+}). Corrosive sublimate is mercuric chloride ($HgCl_2$), a highly toxic compound where mercury is in the +2 oxidation state.

27. B) Mercurous oxide (Hg_2O) | Explanation : Ozone acts as a strong oxidizing agent and oxidizes mercury to mercurous oxide (sub-oxide), Hg_2O. This dissolved oxide lowers the surface tension of the mercury, causing it to leave a 'tail' on the glass surface.

28. B) -NH2 | Explanation : A primary amine has the general formula $R\text{-}NH_2$, where one hydrogen of ammonia is replaced by an alkyl or aryl group. -NH- is secondary, -N= is tertiary, and $-CONH_2$ is an amide.

29. C) C_nH_{2n+2} | Explanation : Alkanes are saturated hydrocarbons with the general formula C_nH_{2n+2}. C_nH_{2n} represents alkenes or cycloalkanes, and C_nH_{2n-2} represents alkynes or dienes.

30. C) 100% pure, anhydrous ethanol | Explanation : Absolute alcohol is 100% pure ethanol, completely free from water. 95.6% ethanol is rectified spirit, and ethanol mixed with methanol is methylated spirit.

31. D) Chloroform | Explanation : Trichloromethane ($CHCl_3$) is commonly known as chloroform, famously used historically as an anesthetic.

32. C) Concentrated HNO_3 and concentrated H_2SO_4 | Explanation : The nitrating mixture consists of concentrated HNO_3 and concentrated H_2SO_4. The sulfuric acid acts as a stronger acid, forcing nitric acid to act as a base to generate the strong electrophile, the nitronium ion (NO^{2+}).

33. B) Chromyl chloride (CrO_2Cl_2) in CS_2 | Explanation : The Etard reaction utilizes chromyl chloride (CrO_2Cl_2) to oxidize the methyl group of toluene into a chromium complex, which upon hydrolysis yields benzaldehyde, preventing over-oxidation to benzoic acid.

34. A) Highly explosive peroxides | Explanation : Ethers undergo slow auto-oxidation in the presence of air and light to form ether hydroperoxides and dialkyl peroxides, which are highly unstable and can explode upon concentration or heating.

35. C) Acetyl chloride | Explanation : Reactivity of acid derivatives towards nucleophilic acyl substitution depends on the leaving group ability. The chloride ion (Cl^-) is the weakest base and best leaving group, making acid halides (acetyl chloride) the most reactive.

36. C) 1,3,5-Trimethylbenzene (Mesitylene) | Explanation : While ethyne (acetylene) polymerizes to form benzene under these conditions, the cyclic polymerization of three propyne molecules yields 1,3,5-trimethylbenzene, commonly known as mesitylene.

37. C) $-NO_2$ | Explanation : The $-NO_2$ (nitro) group is powerfully electron-withdrawing due to the presence of formally charged nitrogen and electronegative oxygen atoms, making its I effect stronger than halogens (-F) or hydroxyl/amino groups.

38. B) 2-Butene | Explanation : Although 1-butanol is a primary alcohol, the initially formed 1° carbocation undergoes a hydride shift to form a more stable 2° carbocation. Elimination then follows Zaitsev's (Saytzeff's) rule, yielding the more substituted 2-butene as the major product.

39. C) The highly acidic mixture protonates the $-NH_2$ group to form the meta-directing anilinium ion. | Explanation : In the strongly acidic nitrating mixture, the basic $-NH_2$ group is protonated to form the $-NH^{3+}$ (anilinium) ion. Unlike $-NH_2$, the positively charged $-NH^{3+}$ group is strongly electron-withdrawing and meta-directing.

40. B) The acidic proton of ethanol reacts with the Grignard reagent to form an alkane. | Explanation : Grignard reagents are immensely strong bases. They will immediately undergo an acid-base reaction with any substance containing an acidic hydrogen (like alcohols, water, or amines) to form the corresponding alkane (R-H) and a magnesium alkoxide salt.

41. C) Carbonyl chloride (Phosgene) | Explanation : Chloroform ($CHCl_3$) reacts with oxygen in the presence of light to form carbonyl chloride ($COCl_2$), commonly known as phosgene, which is highly toxic. This is why chloroform is stored in dark bottles filled to the brim.

42. A) 2-Methylpropene | Explanation : To find the alkene, remove the oxygen atoms from the carbonyl products and join the carbon atoms with a double bond. Joining acetone [$(CH_3)_2C=O$] and formaldehyde [$O=CH_2$] gives $(CH_3)_2C=CH_2$, which is 2-methylpropene (isobutylene).

43. C) Nucleophilic addition followed by elimination of water | Explanation : Ammonia derivatives (like hydroxylamine) react with carbonyl compounds via a two-step mechanism: first, the nucleophilic nitrogen attacks the electrophilic carbonyl carbon (nucleophilic addition), followed by the elimination of a water molecule to form the C=N double bond.

44. C) Maleic acid readily forms a cyclic anhydride, whereas fumaric acid does not. | Explanation : Maleic acid is the cis-isomer; its two carboxyl (-COOH) groups are on the same side of the double bond, making it structurally easy for them to react, lose water, and form a cyclic anhydride. Fumaric acid is the trans-isomer; its -COOH groups are too far apart to form an anhydride without severe, destructive heating to break the double bond.

45. C) K_2CO_3 | Explanation: The Solvay process relies on the precipitation of the bicarbonate intermediate. While sodium bicarbonate ($NaHCO_3$) is sparingly soluble and precipitates out, potassium bicarbonate ($KHCO_3$) is highly soluble in water and cannot be precipitated, making it impossible to produce K_2CO_3 via this method.

46. B) Propane: Explanation: Propanone is the IUPAC name for acetone. Chloretone (a sleep-inducing drug) is formed by the nucleophilic addition reaction of chloroform with acetone in the presence of an alkali.

47. D) Sodium-24 | Explanation: Na-24 is a radioactive isotope of sodium that is injected into the bloodstream as a tracer to detect blood clots or restricted circulation. (I-131 is for the thyroid, P-32 is for leukemia, and Co-60 is for cancer radiation therapy).

48. B) Hydrazine | Explanation: Lassaigne's test for nitrogen relies on the carbon and nitrogen in the organic compound fusing with sodium metal to form Sodium Cyanide (NaCN). Hydrazine (NH_2NH_2) contains nitrogen but lacks carbon, so it cannot form the CN- ion required for the test.

49. B) Isotopes | Explanation: Atmolysis is a separation technique based on Graham's law of diffusion. It is used to separate a mixture of gases (or gaseous isotopes, like Uranium-235 and Uranium-238 as hexafluorides) that have different molecular masses and therefore diffuse at different rates through a porous barrier.

50. D) None. | Explanation: In permanganometric titrations, Potassium Permanganate ($KMnO_4$) acts as a self-indicator. The endpoint is marked by the appearance of a permanent pale pink color when a slight excess of the purple MnO_4- ion is added.

PHYSICS SOLUTIONS

51. C) Work | Explanation : Work is the dot product of two vectors (force and displacement), resulting in a scalar quantity. It has magnitude but no direction. Torque, impulse, and momentum are all vector quantities requiring both magnitude and direction.

52. C) Poisson's ratio | Explanation : Poisson's ratio is the fundamental property of a material defined as the ratio of transverse (lateral) strain to axial (longitudinal) strain when stretched or compressed.

53. B) Bernoulli's principle | Explanation : According to Bernoulli's principle, an increase in the speed of a fluid occurs simultaneously with a decrease in pressure. The wing's shape causes air to travel faster over the top surface, creating lower pressure above and higher pressure below, resulting in an upward lift force.

54. B) 50 m | Explanation : The displacement is equal to the area under the velocity-time graph. The shape is a triangle with base = 10 s and height = 10 m/s. Area = 1/2 × base × height = 1/2 × 10 × 10 = 50 m.

55. C) Zero | Explanation : Work done is the dot product of force and displacement ($W = F \cdot d \cos \theta$). In uniform circular motion, the centripetal force is directed towards the center, while the instantaneous displacement is tangential. Since the angle between them is always 90° (cos 90° = 0), the work done is zero.

56. C) Become one-fourth | Explanation : According to Newton's Law of Universal Gravitation, $F = G(m1 \cdot m2)/r^2$. Force is inversely proportional to the square of the distance. If r becomes 2r, the denominator becomes $(2r)^2 = 4r^2$. The new force is 1/4th of the original force.

57. B) 6 N | Explanation : First, find the common acceleration of the system: a = Total Force / Total Mass = 10 N / (2 kg + 3 kg) = 2 m/s^2. The contact force is the force required to push only the 3 kg block forward at this acceleration. Force on m2 = m2 × a = 3 kg × 2 m/s^2 = 6 N.

58. A) Solid cylinder | Explanation : The acceleration of a rolling body down an incline is $a = g \sin \theta / (1 + K^2/R^2)$, where K is the radius of gyration. A hollow cylinder has its mass concentrated further from the center, so its moment of inertia (and K^2/R^2) is greater. A greater inertia means higher resistance to rotational acceleration. Therefore, the solid cylinder accelerates faster and reaches the bottom first.

59. C) Remain the same | Explanation : According to Archimedes' principle, the floating ice displaces a volume of water whose weight equals the total weight of the ice. When the ice melts, it turns into water with exactly that same weight and therefore fills the exact same volume it previously displaced. Thus, the water level does not change.

60. C) 2T | Explanation : Acceleration due to gravity at a height h is $g' = g / (1 + h/R)^2$. Since h = R, $g' = g / (1 + 1)^2 = g/4$. The time period of a simple pendulum is $T = 2\pi\sqrt{(L/g)}$. The new time period $T' = 2\pi\sqrt{(L/g')} = 2\pi\sqrt{(L/(g/4))} = 2\pi\sqrt{(4L/g)} = 2 \times [2\pi\sqrt{(L/g)}] = 2T$.

61. C) Radiation thermometer (Pyrometer) | Explanation : Radiation thermometers work on the principle of thermal radiation (linked to the Stefan-Boltzmann law). They do not require direct physical contact with the object, making them the only practical choice for measuring extremely high temperatures where other physical thermometers would melt or break.

62. D) Volume | Explanation : By definition, an isochoric process is a thermodynamic process occurring at a constant volume ($\Delta V = 0$). Because volume does not change, the system does no mechanical work.

63. B) Increase by a factor of 2 | Explanation : The rms speed of gas molecules is directly proportional to the square root of its absolute temperature ($v_{rms} \propto \sqrt{T}$). If T becomes 4T, the new velocity is $\sqrt{4}$ = 2 times the original speed.

64. C) 800 cal | Explanation : The heat required for a phase change at constant temperature relies on latent heat, calculated via the formula Q = mL. Here, m = 10 g and L = 80 cal/g. Q = 10 g × 80 cal/g = 800 cal.

65. B) Increase, because thermal expansion acts like a photographic enlargement | Explanation : A common misconception is that the material expands inward to fill the hole. In reality, thermal expansion is isotropic (uniform in all directions). The distance between any two points in the object increases. Therefore, the cavity expands exactly as if it were a solid piece of the same material being heated.

66. C) Increase | Explanation : The efficiency of an ideal (Carnot) engine is $\eta = 1 - (T_C/T_H)$. Lowering T_C decreases the value of the fraction (T_C/T_H). Subtracting a smaller fraction from 1 results in a higher overall efficiency η. Note that it cannot reach 100% unless TC reaches absolute zero (0 K), which is practically impossible.

67. B) -50 J | Explanation : The First Law of Thermodynamics states $\Delta Q = \Delta U + \Delta W$. In an adiabatic process, there is no heat exchange with the surroundings, meaning $\Delta Q = 0$. Therefore, $0 = \Delta U + \Delta W$. Since the gas expands and does work on its surroundings, ΔW is positive (+50 J). Substituting this in yields ΔU = -50 J. The internal energy decreases.

68. C) Polarization | Explanation : Polarization is the restriction of the vibrations of a wave to a single plane. This is only possible for transverse waves (like light), where vibrations are perpendicular to the direction of propagation. Longitudinal waves (like sound) cannot be polarized.

69. A) The pressure of the gas | Explanation : According to Laplace's formula, $v = \sqrt{(\gamma P/\rho)}$. At constant temperature, Boyle's Law states that pressure is directly proportional to density ($P \propto \rho$). Therefore, any change in pressure results in a proportional change in density, keeping the ratio P/ρ constant.

70. C) Four times its original value | Explanation : The fringe width (β) is given by $\beta = (\lambda D)/d$. If D becomes 2D and d becomes d/2, the new fringe width $\beta' = (\lambda * 2D) / (d/2) = 4 * (\lambda D/d) = 4\beta$.

71. D) 80 cm | Explanation : Using the Lens Maker's Formula: $f_{water} / f_{air} = (\mu_g - 1) / ((\mu_g / \mu_w) - 1)$. $f_{water} / 20 = (1.5 - 1) / ((1.5 / 1.33) - 1) = 0.5 / (1.125 - 1) = 0.5 / 0.125 = 4$. Therefore, $f_{water} = 4 * 20 = 80$ cm.

72. C) 25% | Explanation : The Doppler effect formula for a moving source and stationary observer is $f' = f[v / (v - v_s)]$. Here, $v_s = v/5$. $f' = f [v / (v - v/5)] = f [v / (4v/5)] = (5/4)f = 1.25f$. The increase is 0.25f, which corresponds to a 25% increase. (Many incorrectly assume 20% by mixing the formulas for observer motion versus source motion).

73. B) 81:1 | Explanation : Intensity is proportional to the square of amplitude ($I \propto a^2$), so $a1/a2 = \sqrt{25} / \sqrt{16} = 5/4$. $I_{max} / I_{min} = (a_1 + a_2)^2 / (a_1 - a_2)^2 = (5 + 4)^2 / (5 - 4)^2 = 9^2 / 1^2 = 81/1$.

74. C) is perpendicular to the face of emergence | Explanation : Given angle of incidence i = 60°, prism angle A = 30°, and angle of deviation δ = 30°. The relation for a prism is $\delta = i + e - A$. Substituting the values: 30° = 60° + e - 30° => 30° = 30° + e => e = 0°. Since the angle of emergence e = 0°, the emergent ray travels along the normal, meaning it is exactly perpendicular to the face of emergence.

75. A) 3 I_0 / 32 | Explanation : Let I_0 be the initial unpolarized intensity. After the 1st polaroid, intensity $I_1 = I_0 / 2$. The 2nd (middle) polaroid is at 30° to the first. Using Malus's Law: $I_2 = I_1 * \cos^2(30°) = (I_0/ 2) * (\sqrt{3} / 2)^2 = 3 I_0 / 8$. The 3rd polaroid is at 90° to the 1st, so its angle relative to the middle polaroid is 90° - 30° = 60°. Final intensity I_3 = I_2 * $\cos^2(60°) = (3 I_0 / 8) * (1/2)^2 = (3 I_0 / 8) * (1/4) = 3 I_0 / 32$.

76. C) 4F | Explanation : The force between two charges in a medium is $F_{medium} = F_{vaccume} /K$. Therefore, if the initial force in the medium is F, the force in a vacuum (where K=1) will be $F_{vaccume}$ = K * F. Since K=4, the new force is 4F.

77. D) Zero | Explanation : A semicircular path centered perfectly on a point charge represents an equipotential line (since the distance r from the central charge remains constant, the potential V is constant). The work done moving a charge along an equipotential surface is always $W = q\Delta V = 0$.

78. B) $(2K\varepsilon_0 A)/(d(K+1))$ | Explanation : This arrangement can be treated as two capacitors in series: one with the air gap (thickness d/2) and one with the dielectric (thickness d/2). $C_{air} = (\varepsilon_0 A)/(d/2) = 2\varepsilon_0 A/d$. $C_{dielectirc} = (K\varepsilon_0 A)/(d/2) = 2K\varepsilon_0 A/d$. For series: $1/C_{eq} = 1/C_{air} + 1/C_{dielectirc}$. Solving yields $C_{eq} = (2K\varepsilon_0 A)/(d(K+1))$.

79. A) 1.0×10^{-3} J | Explanation : The formula for energy loss when two capacitors are connected in parallel is $\Delta U = 1/2 \cdot [(C_1C_2)/(C_1+C_2)] \cdot (V_1-V_2)^2$. $\Delta U = 0.5 \cdot [(4\times10^{-6} \cdot 1\times10^{-6})/(5\times10^{-6})] \cdot (50-0)^2$. $\Delta U = 0.5 \cdot (0.8\times10^{-6}) \cdot 2500 = 1.0\times10^{-3}$ J.

80. C) 4R | Explanation : Resistance $R = \rho L/A$. If a wire is stretched, its volume ($V=A\cdot L$) remains constant. If length L doubles (L'=2L), the cross-sectional area A must halve (A'=A/2) t
o maintain the volume. $R' = \rho(2L)/(A/2) = 4\cdot(\rho L/A) = 4R$.

81. A) 0 A | Explanation : The thick middle wire connecting the top and bottom rails before the 10 Ω resistor has zero resistance. This creates a "short circuit" completely bypassing the 10 Ω resistor. Current follows the path of least resistance, so all current flows through the short. No current flows through the 10 Ω branch.

82. C) Peltier effect | Explanation : The Peltier effect describes the heating or cooling at an electrified junction of two different conductors. It is the direct converse of the Seebeck effect (which generates a voltage from a temperature difference). Joule heating always evolves heat regardless of direction.

83. B) 2000 W | Explanation : First, find inductive reactance: $X_L = 2\pi fL = 2\pi(50)(0.1/\pi) = 10\ \Omega$. Calculate Impedance: $Z = \sqrt{R^2 + X_L^2} = \sqrt{10^2 + 10^2} = 10\sqrt{2}\ \Omega$. Find RMS Current: $I_{rms} = V/Z = 200/(10\sqrt{2}) = 10\sqrt{2}$ A. Power dissipated only occurs across the resistor: $P = I_{rms}^2 \cdot R = (10\sqrt{2})^2 \cdot 10 = 200 \cdot 10 = 2000$ W.

84. B) Slowly move from the stronger part to the weaker part of the field. | Explanation : A relative permeability slightly less than 1 ($\mu_r < 1$) is the defining characteristic of a diamagnetic material. Diamagnetic materials are weakly repelled by magnetic fields, meaning they experience a force driving them from stronger magnetic field regions toward weaker regions.

85. C) 1/r | Explanation : According to Ampere's Law, the magnetic field produced by an infinitely long straight wire is $B = (\mu_0 I)/(2\pi r)$. Therefore, the magnetic field strength B is inversely proportional to the distance r.

86. B) 1.6×10^{-14} N | Explanation : The magnetic force is given by $F = qvB \sin(\theta)$. $F = (1.6\times10^{-19}\ C) \cdot (1.0\times10^5\ m/s) \cdot (2.0\ T) \cdot \sin(30°)$. Since $\sin(30°) = 0.5$, $F = 1.6\times10^{-19} \cdot 10^5 \cdot 2.0 \cdot 0.5 = 1.6\times10^{-14}$ N.

87. C) Less than gravity (a<g) for the entire duration of the fall. | Explanation : According to Lenz's Law, the induced current in the copper ring will create a magnetic field that opposes the change producing it. As the magnet falls in, the ring repels it (upward force). As it falls out, the ring attracts it (upward force). In both cases, the net downward force is less than gravity, so a<g.

88. A) 0.1 J | Explanation : The energy stored in an inductor is completely independent of the time it took to reach the current, it solely depends on the final current value. $E = 1/2\ LI^2$. $E = 0.5 \cdot (50\times10^{-3}\ H) \cdot (2.0\ A)^2 = 0.5 \cdot 0.05 \cdot 4 = 0.1$ J.

89. C) Nuclear density | Explanation : In nuclear physics, the radius R of a nucleus is related to its mass number A by $R = R_0A^{1/3}$. Since mass $\propto$ A and volume $\propto R^3 \propto A$, nuclear density remains approximately constant for all nuclei.

90. D) Photoelectric emission will not occur | Explanation : The incident photon energy is less than the work function, so electrons cannot escape from the metal surface.

91. B) Directly proportional to its distance from the observer. | Explanation : Hubble's law states $v = H_0d$, so recessional velocity is directly proportional to distance.

92. B) 60 days | Explanation : If 7/8 disintegrates, 1/8 remains. Since $(1/2)^3 = 1/8$, three half-lives are required. $3 \times 20 = 60$ days.

93. A) Positive terminal of battery is connected to p-type and negative to n-type. | Explanation : Forward bias reduces the potential barrier and allows current to flow easily through the junction.

94. C) Specific charge of the electron (e/m) | Explanation : Thomson's experiment determined the charge-to-mass ratio (e/m) of the electron.

95. B) Transitions of inner-shell electrons in the target atoms. | Explanation : Characteristic X-rays are emitted when electrons transition between inner atomic energy levels.

96. C) An XOR (Exclusive OR) structure built from fundamental gates. | Explanation : An XOR gate produces HIGH output only when the two inputs are different.

97. D) It remains unchanged. | Explanation : Maximum kinetic energy depends on frequency, not intensity. Intensity only affects the number of emitted electrons.

98. B) 1 - 1/e | Explanation : After one mean life, the remaining fraction is 1/e. Therefore the disintegrated fraction is $1 - 1/e$.

99. D) $\sqrt{(eV/2mc^2)}$ | Explanation : Using de Broglie wavelength and photon energy relations, the ratio simplifies to $\sqrt{(eV/2mc^2)}$.

100. B) 92.16 MeV | Explanation : Mass defect × 931.5 MeV/amu gives a total binding energy of about 92.16 MeV.

ZOOLOGY SOLUTIONS

101. B) Ancestral giraffes continuously stretched their necks to reach high leaves, and this acquired elongation was passed directly to their offspring. | Explanation : Lamarck's theory centered on the 'use and disuse of organs' and the 'inheritance of acquired characteristics' the idea that physical changes acquired during an organism's lifetime could be inherited.

102. B) Reptilia and Aves (Birds) | Explanation : Archaeopteryx possessed reptilian features (a long bony tail, teeth in its beak) but also possessed fully formed feathers and wings, proving that birds evolved from theropod dinosaurs (reptiles).

103. D) Neanderthal man (Homo neanderthalensis) | Explanation : Neanderthals had a heavily built skull and a very large brain, averaging around 1450-1600 cc, slightly larger than the modern human average of 1350-1400 cc.

104. B) Nematocyst | Explanation : The cell itself is the cnidocyte (or cnidoblast), but the complex, harpoon-like organelle inside it that is ejected under osmotic pressure is the nematocyst.

105. B) The presence of a tough, enzyme-resistant syncytial tegument and total loss of the alimentary canal. | Explanation : Tapeworms absorb pre-digested nutrients directly across their tough tegument. Since they live bathed in host food, they have entirely lost their own digestive tract.

106. C) Cephalopoda (Squids and Octopuses) | Explanation : Cephalopods are predatory and highly active. To meet their high metabolic demands, they possess a closed circulatory system with systemic and branchial (gill) hearts.

107. C) Crocodilia (Crocodiles) | Explanation : The description defines reptiles. Crocodilians (crocodiles, alligators) are unique among reptiles because their ventricle is completely divided by a septum, giving them a 4-chambered heart like birds and mammals.

108. C) Brown adipose tissue | Explanation : Brown fat is highly vascularized and packed with mitochondria. Thermogenin uncouples the electron transport chain from ATP synthesis, causing the energy to be released purely as heat.

109. B) Intramembranous ossification | Explanation : Intramembranous ossification is the direct conversion of mesenchymal tissue into bone, forming the flat bones of the skull and the clavicle. Long bones use endochondral ossification (replacing a hyaline cartilage model).

110. C) The presence of distinct, regular transverse striations (A and I bands). | Explanation : Skeletal muscle fibers are highly organized into regular sarcomeres, creating dark (A) and light (I) bands (striations). Smooth muscle lacks this regular arrangement, appearing unstriated.

111. C) Nodes of Ranvier | Explanation : The Nodes of Ranvier are the naked gaps between adjacent Schwann cells. They have a massive density of ion channels, allowing the action potential to be regenerated at each node.

112. C) Blackwater fever | Explanation : Blackwater fever is an autoimmune-like reaction triggered by P. falciparum, causing massive lysis of both infected and uninfected RBCs. The spilled hemoglobin turns the urine dark/black.

113. A) Oocyst | Explanation : The ookinete forms an oocyst on the hemocoel side of the mosquito's gut wall. Inside the oocyst, thousands of sporozoites develop, which eventually burst out and migrate to the salivary glands.

114. C) Segment 26 | Explanation : The typhlosolar region of the earthworm intestine begins exactly at the 26th segment and continues up to the last 23-25 segments of the body, vastly increasing the absorptive surface area.

115. B) Segments 10 and 11 | Explanation : In the earthworm, the male reproductive system features two pairs of testes enclosed in testis sacs located ventrally in segments 10 and 11.

116. D) Columella auris | Explanation : Amphibians have a single middle ear bone (ossicle), the columella auris, which transmits vibrations directly from the surface tympanum to the oval window of the inner ear.

117. B) 10 pairs | Explanation : Unlike amniotes (reptiles, birds, mammals) which have 12 pairs of cranial nerves, amphibians (like the frog) and fish possess exactly 10 pairs of cranial nerves.

118. C) Inducing vigorous contraction of the gallbladder and stimulating the pancreas to release enzyme-rich juice. | Explanation : CCK gets its name ('chole'=bile, 'cyst'=sac, 'kinin' move) because it forces the gallbladder to contract, dumping bile into the duodenum to emulsify the fats that triggered its release. It also prompts the pancreas to release digestive enzymes.

119. D) Terminal ileum | Explanation : The intrinsic factor-B12 complex travels completely intact through the GI tract until it reaches the terminal ileum, where specific receptors (cubilin) mediate its endocytosis into the blood.

120. B) Shift extremely to the left, preventing the release of any bound oxygen to the tissues. | Explanation : CO not only occupies oxygen-binding sites but also locks the remaining oxygen molecules onto the hemoglobin tetramer (left shift). Thus, the blood cannot unload what little oxygen it has into the hypoxic tissues.

121. B) Hering-Breuer reflex | Explanation : The Hering-Breuer inflation reflex acts as a protective mechanism. When the lungs inflate excessively, vagal afferents inhibit the dorsal respiratory group (DRG), forcing expiration to begin.

122. C) Isovolumetric relaxation | Explanation : Isovolumetric relaxation occurs during early diastole. The aortic valve snaps shut (dub sound), but the pressure is still too high for the mitral valve to open. Volume is constant, but pressure plummets.

123. A) Blood Colloid Osmotic Pressure (Oncotic pressure) | Explanation : Oncotic pressure is the osmotic pressure exerted by large plasma proteins that cannot easily cross the capillary wall. This 'pull' is essential for reclaiming fluid from the tissue spaces to prevent edema.

124. C) Glomerulus | Explanation : The glomerulus is unique in the circulatory system because it is fed and drained by arterioles, maintaining a very high hydrostatic pressure necessary to force plasma out into the Bowman's capsule.

125. B) Secrete more hydrogen ions (H+) into the urine and reabsorb new bicarbonate. | Explanation : To buffer the excess acid (low pH), the kidneys upregulate H+ secretion (via intercalated cells) and synthesize/reabsorb more bicarbonate to return the blood pH to normal (metabolic compensation).

126. C) Voltage-gated sodium Na^+ channels | Explanation : Local anesthetics block voltage-gated Na^+ channels. Without Na^+ influx, depolarization cannot occur, effectively halting the generation and propagation of pain action potentials to the brain.

127. A) Left frontal lobe | Explanation : Broca's area controls the motor mechanics of speech and is overwhelmingly lateralized to the left frontal lobe. Damage here causes expressive (non-fluent) aphasia-understanding is intact, but speaking is nearly impossible.

128. B) Fovea centralis | Explanation : The fovea centralis lacks rod cells and blood vessels, allowing light to strike the densely packed cones directly, providing the highest resolution central vision.

129. B) Aldosterone (Mineralocorticoid) | Explanation : The adrenal cortex has three zones: Glomerulosa (Aldosterone for salt/water balance), Fasciculata (Cortisol for stress/metabolism), and Reticularis (Androgens).

130. C) Growth Hormone (GH) | Explanation : Somatostatin is also known as Growth Hormone-Inhibiting Hormone (GHIH). It directly blocks the somatotrophs in the anterior pituitary from releasing Growth Hormone.

131. B) Lysosome / Golgi complex | Explanation : The acrosome forms during spermiogenesis from the Golgi apparatus. It acts effectively as a giant, specialized lysosome, packed with the enzymes needed to digest the zona pellucida.

132. A) Trophoblast | Explanation : The trophoblast provides nutrients to the embryo, secretes hCG to maintain the pregnancy, and eventually develops into the chorion and the fetal contribution to the placenta.

133. B) Granuloma (Tubercle) | Explanation : When macrophages cannot kill Mycobacterium tuberculosis, they, along with T-cells, form a tightly packed sphere (granuloma or tubercle) around the bacteria to contain the infection, leading to central tissue death (caseation).

134. B) Double-stranded DNA virus | Explanation : Hepatitis A, C, D, and E are all RNA viruses. Hepatitis B (HBV) is a hepadnavirus, which is a partially double-stranded DNA virus that replicates through an RNA intermediate using reverse transcriptase.

135. B) Coating of a pathogen with antibodies or complement proteins to mark it for rapid phagocytosis. | Explanation : Opsonins (like IgG antibodies and C3b) act like handles or 'ketchup'. Macrophages and neutrophils have receptors for these opsonins, making it much easier for them to grab and devour the slippery bacteria.

136. B) IgG antibodies | Explanation : The primary response produces predominantly IgM. The secondary (memory/booster) response produces a massive, rapid surge of high-affinity IgG antibodies due to the activation of long lived memory B-cells.

137. C) Lactobacillus acidophilus | Explanation : Lactobacillus (Lactic Acid Bacteria or LAB) converts lactose sugars into lactic acid. The resulting acidic environment drops the pH, denaturing and precipitating the casein proteins to form yogurt or cheese curds.

138. B) Selectively disabling or disrupting a specific existing gene to observe the resulting phenotypic defect. | Explanation : Gene knockout involves replacing a functional gene with an inactive, mutant sequence. By observing what goes wrong in the
'knockout' mouse, researchers can deduce the exact physiological function of that specific gene.

139. C) Methylmercury | Explanation : Industrial wastewater containing mercury was discharged into Minamata Bay. Bacteria converted it to highly toxic methylmercury, which bioaccumulated in shellfish and fish, causing extreme neurological damage to humans who ate them.

140. B) The Snow Leopard (Panthera uncia) | Explanation : Shey Phoksundo is Nepal's largest national park, located in the trans-Himalayan region. Its harsh, high-altitude rocky terrain is the prime, natural habitat for the endangered Snow Leopard and its primary prey, the blue sheep.

BOTANY SOLUTIONS

141. C) Lowering the activation energy required for biochemical reactions | Explanation : Enzymes are biological catalysts (mostly proteins) that speed up metabolic reactions by lowering the activation energy.

142. B) The absence of double bonds between carbon atoms in its fatty acid chains | Explanation : Saturated fatty acids lack double bonds, allowing them to pack closely together, making the lipid solid at room temperature.

143. C) Mucor - Fungi | Explanation : Mucor is a fungus (Zygomycete) and belongs to the Kingdom Fungi. Cyanobacteria are Monera, Yeast is Fungi, and Spirogyra (Algae) can be Plantae, but Mucor to Fungi is a textbook definitive match.

144. B) Cyanobacteria | Explanation : Many cyanobacteria (like Anabaena and Nostoc) can fix atmospheric nitrogen, greatly enriching soil fertility, especially in aquatic environments like rice paddies.

145. A) A(9)+1 | Explanation : Fabaceae exhibits diadelphous stamens, typically where 9 stamens are fused together and 1 is free, represented as A(9)+1.

146. C) Marchantia | Explanation: Gemmae develop inside these cups and detach to grow into new plants. Therefore, Marchantia bears gemma cups for asexual reproduction.

147. B) Ocimum sanctum

148. C) Gametophyte | Explanation**:** The gametophyte is the dominant, photosynthetic, and independent generation in *Marchantia*, whereas the sporophyte remains attached to it.

149. C) Xerophytes | Explanation**:** In xerophytes, leaves are often reduced to spines to minimize water loss.
The stem becomes green and fleshy (phylloclade/cladode) to carry out photosynthesis and store water.

150. B) Embryo and endosperm | Explanation**:** One male gamete fuses with the egg to form the embryo, while the other fuses with the polar nuclei to form the endosperm.

151. C) Carolus Linnaeus | Explanation: Carolus Linnaeus developed the system of binomial nomenclature, assigning organisms a genus and species name.

152. B) Zooplankton | Explanation**:** Zooplankton feed on phytoplankton and occupy the primary consumer level in the pond food chain.

153. B) Floating plant stage | Explanation: In a hydrosere, succession progresses from phytoplankton -> submerged -> floating -> reed-swamp -> marsh-meadow -> woodland -> climax forest.

154. C) Absorb and re-radiate long-wave infrared radiation back to Earth | Explanation : Greenhouse gases let short-wave solar radiation pass through but absorb the long-wave infrared radiation emitted by the Earth, re-radiating it back and warming the planet.

155. A) Ozone + PAN + NO_x | Explanation: Photochemical smog is a brownish smog commonly seen in urban areas with heavy vehicle emissions. Its major components are ozone (O_3), PAN, and nitrogen oxides (NOx).

156. C) Golgi apparatus | Explanation**:** Lysosomes originate from the Golgi apparatus and contain digestive enzymes used for intracellular digestion.

157. C) Metaphase | Explanation: Spindle fibers are essential for aligning chromosomes at the metaphase plate and separating them during anaphase. Blocking them arrests the cell in metaphase (e.g., colchicine action).

158. C) Ribosome | Explanation**:** Ribosomes occur in both prokaryotic and eukaryotic cells and are responsible for protein synthesis.

159. C) Repair of damaged somatic tissues and vegetative growth | Explanation: Tissue repair and somatic growth are functions of mitosis, not meiosis, which is restricted to gamete formation.

160. B) A direct cell division without the appearance of chromosomes or spindle fibers | Explanation : Amitosis is a simple, direct method of cell division involving the elongation and constriction of the nucleus without the complex phases seen in mitosis.

161. C) X-linked recessive inheritance | Explanation**:** The gene for color blindness is located on the X chromosome and is inherited as a recessive trait.

162. B) 1:2:1 | Explanation : In incomplete dominance, the heterozygous phenotype is intermediate, making the phenotypic ratio (1 Red: 2 Pink: 1 White) identical to the genotypic ratio

163. B) 45, XO | Explanation : Turner's syndrome occurs in females missing one X chromosome, resulting in a 45, XO karyotype.

164. B) DNA to RNA to Protein | Explanation : The central dogma describes the flow of genetic information: DNA is transcribed into RNA, which is translated into Protein.

165. C) Albinism | Explanation : Albinism is an autosomal recessive disorder. Hemophilia and color blindness are X-linked recessive. Down's syndrome is a chromosomal abnormality (trisomy 21).

166. A) Colchicine | Explanation : Colchicine is an alkaloid that prevents microtubule polymerization, stopping spindle formation and doubling the chromosome number (inducing polyploidy) in treated cells.

167. B) Radial and exarch | Explanation : Roots typically have radial vascular bundles where xylem and phloem are on different radii. In roots, protoxylem is towards the periphery (exarch arrangement).

168. C) Lateral meristem (Cambium) | Explanation : Lateral meristems, which include the vascular cambium and cork cambium, divide laterally to add secondary tissues, increasing the stem's girth.

169. B) Monocot stem | **Explanation :** Closed vascular bundles lack cambium and therefore do not undergo secondary growth. They are characteristic of monocot stems.

170. B) Water will move from the cell into the solution | Explanation : Cell Water Potential = Osmotic Potential + Turgor Pressure = -10 + 4 = -6 bars. Water moves from higher potential (-6 bars) to lower potential (-8 bars), hence out of the cell.

171. C) 3-Phosphoglyceric acid (3-PGA) | Explanation : In the C3 cycle, the enzyme RuBisCO fixes CO_2 to RuBP, immediately splitting it into two molecules of the 3-carbon compound 3-PGA.

172. C) The cohesive and adhesive properties of water molecules | Explanation : The cohesion-tension theory explains that water molecules stick to each other (cohesion) and to the xylem walls (adhesion), allowing transpiration pull to draw water upward.

173. C) Auxin | Explanation : Auxins, produced in the apical meristem, travel downwards and inhibit the growth of lateral (axillary) buds, a phenomenon known as apical dominance.

174. A) Mesophyll cells | Explanation : In C4 plants, initial CO_2 fixation occurs in the mesophyll cells via PEP carboxylase, forming a 4-carbon acid which is then transported to the bundle sheath cells for the Calvin cycle.

175. C) Guttation | Explanation : Guttation occurs typically at night or early morning under conditions of high soil moisture and low transpiration, driven by positive root pressure.

176. B) One male gamete and the diploid secondary nucleus (two polar nuclei) | Explanation : During double fertilization, one male gamete fertilizes the egg (forming the zygote), and the other fuses with the two polar nuclei to form the triploid primary endosperm nucleus.

177. C) Autogamy | Explanation : Autogamy is self-pollination occurring within a single flower. Geitonogamy is between different flowers on the same plant, and xenogamy is between different plants.

178. B) The inherent capacity of a single somatic plant cell to divide and differentiate into a complete whole plant | Explanation : Totipotency is the genetic potential of a single living plant cell, given the right nutrients and hormones, to regenerate into an entire functional plant.

179. C) Biofortification | Explanation : Biofortification is the process of improving the nutritional quality of food crops through agronomic practices, conventional plant breeding, or modern biotechnology.

180. C) Azolla in symbiotic association with Anabaena | Explanation : The aquatic fern Azolla harbors the nitrogen-fixing cyanobacterium Anabaena in its leaves. It is heavily used in rice cultivation to naturally increase nitrogen levels in flooded fields.

MAT SOLUTIONS

181. C) Eating | Explanation: A book is used for reading, just as a fork is used for eating.

182. B) JNRGH | Explanation: Each letter is shifted one position backward in the alphabet.

183. C) His son's | Explanation: 'My father's son' refers to Ram himself, so the man is Ram's son.

184. A) East | Explanation: Following the turns step by step leaves the person facing East.

185. B) B | Explanation: Doctors and Lawyers are separate subsets within Professionals.

186. C) 5.6 | Explanation: The first five primes are 2, 3, 5, 7, and 11; average = 5.6.

187. A) 4% of a | Explanation: Substituting b = 0.2a gives b% of 20 = 0.04a.

188. C) 4% Loss | Explanation: Equal gain and loss percentages on equal SP result in a net loss.

189. B) 24 | Explanation: Choose an even last digit and arrange remaining digits.

190. B) 10 days | Explanation: A 50% efficiency increase reduces the time to 10 days.

191. B) The conclusion is False | Explanation: No definite relation between cats and birds can be concluded.

192. C) Both I and II | Explanation: The advertisement assumes both interest and good returns.

193. A) Event 1 causes Event 2 | Explanation: Higher prices generally reduce demand.

194. A) A | Explanation: The dot continues moving clockwise to the next corner.

195. B) Only II | Explanation: Supplying water and fodder is the practical response.

196. B) B | Explanation: A mirror image reverses the figure horizontally.

197. C) C | Explanation: Figure C differs from the others in its internal regions.

198. C) C | Explanation: The final figure is formed by combining previous patterns.

199. A) A | Explanation: A water image is an upside-down reflection.

200. A) A | Explanation: Folding superimposes the marked symbols accordingly.